Non Constructible Change

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The problem

- Given positive coin values, find the minimum amount that cannot be formed.
- Sort; if the next coin exceeds (reachable + 1), reachable + 1 is the answer.

Intuition

- Sort the coins.
- If you can already build every amount up to some total, a coin bigger than total plus one leaves a gap you can never fill.

Approach

- 1. Sort the coins ascending
- 2. Track change, the largest amount you can build so far
- 3. If a coin exceeds change plus one, return change plus one
- 4. Otherwise add the coin to change

Java solution

```java
public class NonConstructibleChange {
    public static int nonConstructibleChange(int[] coins) {
        Arrays.sort(coins);
        int reach = 0;
        for (int c : coins) {
            if (c > reach + 1) break;
            reach += c;
        }
        return reach + 1;
    }
}
```

Complexity

- Time: O(n log n)
- Space: O(1) (in-place sort)

Recap

- Sort the coins.
- Complexity: O(n log n)